Roll No. | | | | | | | |

Paper Code: TCS 302

END SEMESTER Examination 2025

B. Tech (CSE-Core-All) III Sem

Data Structure with 'C' language.

Time: Three Hours | Maximum Marks :100

NSTRUCTIONS TO STUDENTS

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

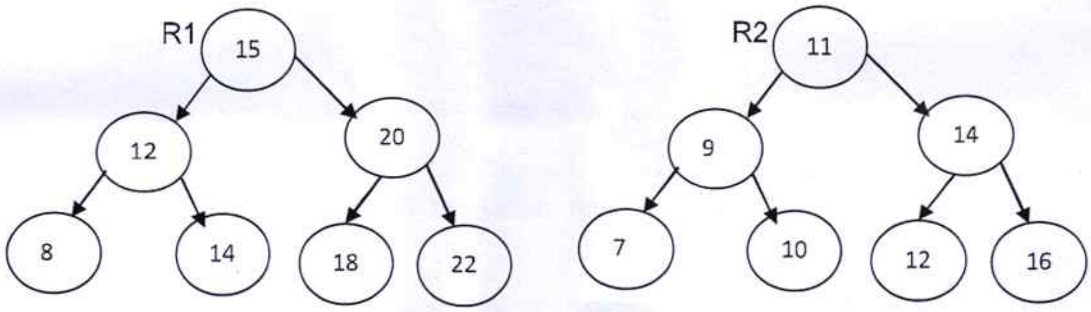
(iii) Each question carries ten marks.

Q1. (2X10=20 Marks) (CO1, CO3, CO4)

A. Assume that we have a single linked list and a key value. The first node of linked list is pointed by a pointer P. Write a C function to delete the last occurrence of that key in the linked list.

B. Explain liner probing and quadratic probing. Consider a hash table of size (m) 13. Using quadratic probing technique with remainder division method (h(key) = key mod table size), insert following keys 13,26,62,83,97,12,65 and 45 into the hash table, where c1=1and c2=2.

C. Assume that you have two binary search trees with pointers R1 and R2, pointing to the roots of the trees. Write a C function to count the number of pairs from the two BSTs whose sum is equal to a given value x.
Example. Input x= 24



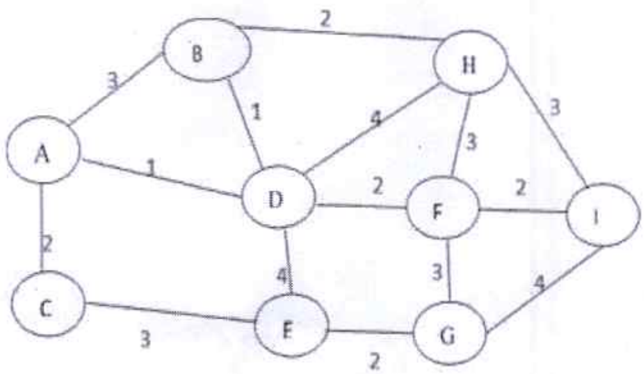
Output: 4 pairs.

Q2. (2X10=20 Marks) (CO2, CO3, CO5)

A. Convert the following infix expression into postfix expression using stack (show all steps).
(A * B) + (C – D) / E % F ^ G

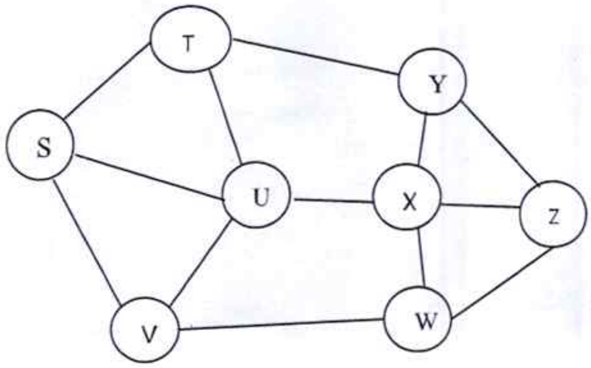
B. Write a C function that concatenates two circulars singly linked lists, ensuring the result is also circular list. Also handle cases where one of the lists is empty.

C. You have a weighted graph for the following situation: A power distribution company wants to connect a set of power stations and substations. The cost of installing transmission lines between any two stations is assigned as weight of the edge. Identify the most suitable algorithm and describe how it helps in constructing a minimum-cost power grid.



Q3. (2X10=20 Marks) (CO2, CO3, CO5)

A. Which graph traversal technique should be used to find one of the possible paths in the following graph from source S to destination Z? Apply the technique on following graph. (Show all steps)



B. Write a C program to delete the minimum element form min heap.

C. Apply merge sort technique to sort the following data, show all the intermediate steps (Do not write code). 8,22,13,4,5,90,60,30,25.

Q4. (2X10=20 Marks) (CO3, CO4, CO5)

A. Construct a balanced multi-way search tree of order 3. Insert the following keys in the given order: 8,2,14,15,4,5,25,6,1,30,7.

B. Assume that we have a single linked list and value of group(K). The first node of linked list is pointed by a pointer P. Write a C function that reverses every alternate group of k nodes in a singly linked list.

Example:

Input: 10→20→30→40→50→60→70→80

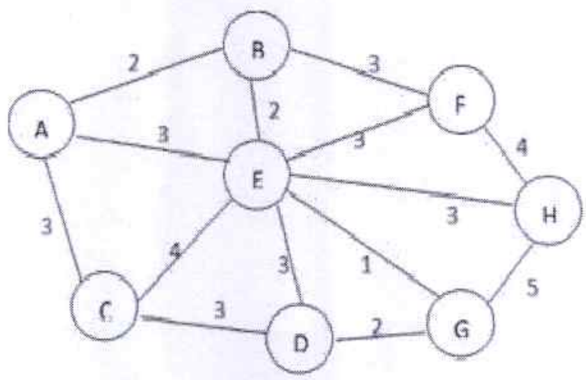
k=2

Output: 20 → 10 → 30 → 40 → 60 → 50 → 70 → 80

C. Which algorithm would you use to compress data by assigning shorter codes to more frequent items? Find the minimum weighted path length for the given data using that algorithm. aabcdefedfcdd.

Q5. (2X10=20 Marks) (CO3, CO4, CO5)

A. A navigation system needs to calculate the shortest driving distance from a user's current location to all other important landmarks in a city. The city map is given as a weighted graph where weights represent distances in kilometers. Identify the algorithm used for this purpose and explain how it finds the shortest path to each destination (Consider vertex A as the source vertex).



B. Write down an algorithm to add all elements of two 2-D arrays into third 2-D array and then predict the nature of algorithm in terms of Big-Oh notation.

C. Write a C program to create a weighted graph with the help of node pointers and linked list of edges.